

CS 593: Big Data and Map Reduce

Course Code: CS 593

Course Name: Big Data and Map Reduce

L-T-P-C: 3-0-0-3

Approval: One-time approval (OTA), recommended by the 26th BoA and approved in the 18th Senate (CS 561 Map Reduce and Big Data was approved in the 9th Senate)

Course Outline

This course aims to provide the students with a thorough understanding of the Map Reduce paradigm and its application to Big Data problems. The course will provide a hands-on programming experience and teach the student to think in terms of Map and Reduce to solve any kind of large scale problem that deals with huge data and needs distributed fault-tolerant computation.

Unit 1 and 2 will be delivered through lectures while Unit 3 will be delivered by a combination of lectures, student presentations and programming projects/demonstrations.

Modules

- **Unit 1:** Introduction to Big Data, the Map Reduce paradigm and programming model. The Map Reduce framework and its benefits. Cloud computing and Map Reduce. Open source Map Reduce frameworks – Hadoop, Shark, Mrjob etc. – comparison and benefits of each such framework. (3 hrs)
- **Unit 2:** Thinking in the Map Reduce way via simple problems that serve as a building block for larger problems – Matrix-Vector multiplication, Matrix multiplication, Relational algebra – selections, projections, union, intersection, difference, natural join, grouping and aggregation. Complexity analysis of Map Reduce algorithms. (3 hrs)
- **Unit 3: Applying Map Reduce to different Big Data contemporary problem areas.** (36 hrs)
 - Similarity computations
 - Clustering algorithms – K-means, CURE, ...
 - Web crawling and indexing
 - Web-scale graph algorithms – PageRank, HITS, ...
 - Recommendation Systems – content-based filtering, collaborative filtering, dimensionality reduction.
 - Text advertising on the web – AdWords.
 - Social networks and their analysis.
 - Large-scale Machine Learning – Perceptrons and Support Vector machines.
 - Statistical Machine Translation.
 - Market Basket analysis.
 - Mining Data Streams